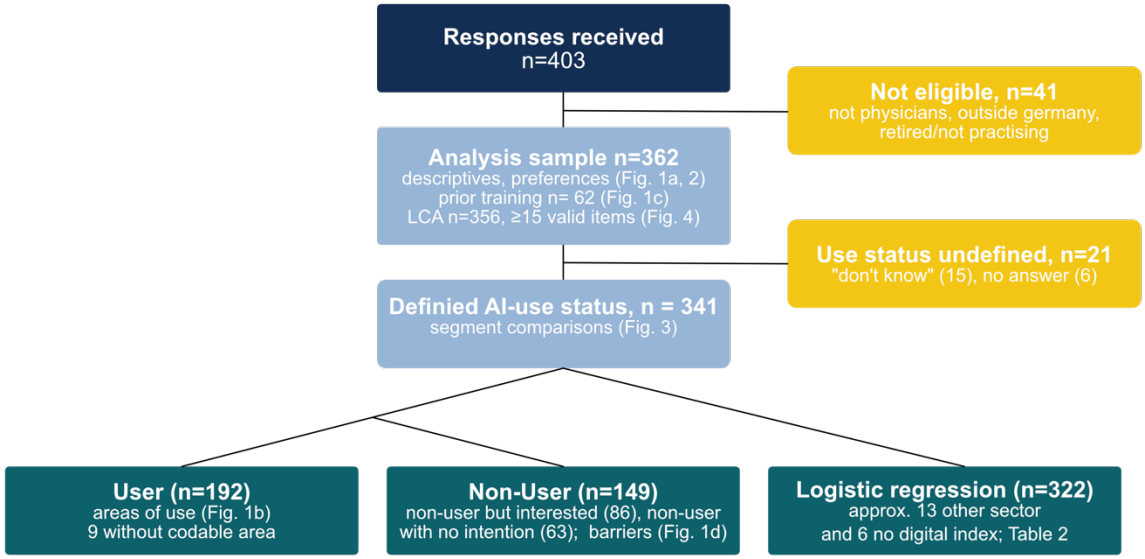
Supplements

Supplements for the Publication "*Consensus on delivery, divergence on content: physicians' artificial intelligence training needs in continuing medical education.*"

Supplementary Note 1

STROBE flowchart and explanation for exclusion.



Data cleaning was performed on the raw sample of n = 403 responses. No duplicate entries were identified. Respondents were excluded stepwise against predefined eligibility criteria (see table below): respondents who were not physicians (other healthcare professionals, nursing staff, medical assistants [MFA], or medical students; n = 9, with free-text and inconsistent job titles standardised to "Ärzt:in"); respondents practising outside Germany (Austria 1, Switzerland 3; n = 4); respondents who were retired or not working in a clinical environment (retirement, self-governance/politics, administration/management, and individual cases indicating no active clinical practice; n = 24); and respondents with missing values in key demographic variables (federal state 2, age 1, sex 1; n = 4). This yielded the final analytic sample of n = 362, comprising physicians from all 16 federal states.

In a second pass, a logical-consistency check of the employer-provision items identified three records in which respondents reported "no tools available" or use of freely available ("shadow") AI while simultaneously indicating a form of employer provision (#89, #345, #316). These records were recoded to "no employer provision" and the corresponding "form of provision" field was cleared, following predefined rules (codebook, Supplementary Note 4). They were corrected, not excluded, and remain in the analytic sample (dataAfterCleaning.csv).

Reason for not eligibility	n	Original Data
No clinician (other occupation, MFA, Nurse oR Student)	9	Column M
Working outside Germany (Austria 1, Switzerland 3)	4	Column J
Retirement (5) or not working in clinical environment (incl. Politics 5, Administration 5, Individual Cases 10)	24	Column J, L, AN, BE
Missing Values (state 2, age 1, sex 1)	4	Column J, I, K

Supplementary Note 2

Characteristic	n (%)
Age group	
20–29 years	27 (7.5)
30–39 years	104 (28.7)
40–49 years	60 (16.6)
50–59 years	86 (23.8)
60–69 years	66 (18.2)
≥70 years	19 (5.2)
Sex	
Female	183 (50.6)
Male	179 (49.4)
Work setting	
Ambulatory care – solo practice	77 (21.3)
Ambulatory care – group practice	85 (23.5)
Ambulatory care – medical care centre (MVZ)	33 (9.1)
Hospital – inpatient	136 (37.6)
Hospital – outpatient	16 (4.4)
Public health service	5 (1.4)
Emergency medical service	5 (1.4)
No information	5 (1.4)
Professional experience (since licensure)	
0–4 years	65 (18.0)
5–9 years	68 (18.8)
10–14 years	33 (9.1)
15–19 years	19 (5.2)
20–25 years	30 (8.3)
>25 years	145 (40.1)
No information	2 (0.6)
Self-rated professional digital competence	
1 (very low)	11 (3.0)
2	65 (18.0)
3	141 (39.0)
4	100 (27.6)
5 (very high)	45 (12.4)
Mean ± SD	3.28 ± 1.00

Professional experience was assessed from the third state examination (licensure). One respondent reported experience only in the item for non-physician staff (counted under >25 years); 'no information' refers to explicit non-response (n = 2). Digital competence was self-rated in the professional context on a 5-point scale (1 = very low, 5 = very high).

Federal state of practice (N = 362). Values are n (%).

Characteristic	n (%)
Nordrhein-Westfalen	65 (18.0)
Bayern	62 (17.1)
Baden-Württemberg	57 (15.7)
Niedersachsen	32 (8.8)
Sachsen	25 (6.9)
Rheinland-Pfalz	22 (6.1)
Berlin	21 (5.8)
Hessen	20 (5.5)
Schleswig-Holstein	15 (4.1)
Brandenburg	13 (3.6)
Hamburg	11 (3.0)
Mecklenburg-Vorpommern	5 (1.4)
Sachsen-Anhalt	5 (1.4)
Thüringen	5 (1.4)
Saarland	3 (0.8)
Bremen	1 (0.3)

Medical specialty (N = 362). Values are n (%). Categories as reported by participants.

Characteristic	n (%)
General practice	70 (19.3)
Internal medicine	41 (11.3)
Paediatrics	27 (7.5)
Gynaecology and obstetrics	27 (7.5)
Orthopaedics and trauma surgery (incl. plastic/reconstructive surgery)	26 (7.2)
Psychiatry	21 (5.8)
Anaesthesiology	17 (4.7)
Neurology	13 (3.6)
Ophthalmology	12 (3.3)
Cardiology	11 (3.0)
Radiology	11 (3.0)
Dermatology	9 (2.5)
General surgery	9 (2.5)
Psychosomatic medicine	8 (2.2)
Otorhinolaryngology	7 (1.9)
Child and adolescent psychiatry	6 (1.7)
Occupational medicine	5 (1.4)
Cardiothoracic and vascular surgery	4 (1.1)
Urology	4 (1.1)
Radiation therapy	3 (0.8)

Geriatrics	3 (0.8)
Laboratory medicine	3 (0.8)
Phoniatrics and paediatric audiology	2 (0.6)
Oral and maxillofacial surgery	2 (0.6)
Neurosurgery	2 (0.6)
Pathology	2 (0.6)
Gastroenterology	2 (0.6)
Palliative medicine	2 (0.6)
Transfusion medicine	2 (0.6)
No information	2 (0.6)
Microbiology, virology and infection epidemiology	1 (0.3)
Rehabilitation medicine	1 (0.3)
Nephrology	1 (0.3)
Orthopaedics	1 (0.3)
Physical and rehabilitative medicine	1 (0.3)
Public health	1 (0.3)
Intensive care medicine	1 (0.3)
Dentistry	1 (0.3)
Others	1 (0.3)

Comparison of the study sample with the German physician population (Bundesärztekammer physician statistics, 31 December 2025).

Characteristic	Study sample (N = 362), %	German physicians (BÄK 2025, N = 446,120), %
Age group		
<40 years	36.2	33.1
40–49 years	16.6	23.3
50–59 years	23.8	20.2
≥60 years	23.5	23.4
Sex		
Female	50.6	50.5
Male	49.4	49.5
Care setting		
Ambulatory	53.9	39.0
Hospital (inpatient and outpatient)	42.0	52.3
Other / no information	4.1	8.7

BÄK reference population: all professionally active physicians in Germany (N = 446,120). BÄK age groups (≤34, 35–39, 40–49, 50–59, 60–65, >65 years) were collapsed to match the survey categories. BÄK care settings: ambulatory (n = 173,900), inpatient (n = 233,500), public authorities/corporations and other areas (n = 38,700). Survey 'ambulatory' comprises solo practice, group practice and MVZ; survey 'hospital' comprises inpatient and hospital-based outpatient care. Source: Bundesärztekammer, Ärztestatistik zum 31. Dezember 2025.

Supplementary Note 3

Sensitivity Analysis for Multivariable logistic regression of professional AI use (n = 322) in RQ2. Addition to Tab. 2.

Analysis	Employer AI provision OR (95% CI)	Digital maturity OR (95% CI)	Digital competence OR (95% CI)
Main model	11.38 (5.24–24.74)	1.48 (1.09–2.00)	1.30 (0.98–1.72)
Binary employer provision	11.46 (5.31–24.73)	1.48 (1.09–2.00)	1.30 (0.98–1.72)
Excluding solo practices	13.45 (5.15–35.13)	1.62 (1.12–2.33)	1.50 (1.08–2.10)
Sector-neutral index	11.83 (5.44–25.71)	1.45 (1.09–1.93)	1.32 (1.00–1.75)
Within-sector z-standardisation	11.45	1.48 (1.09–2.00)	—

Description of the variables used in the multivariable logistic regression.

Variable	Definition	Coding
Professional AI use	Use of AI tools in clinical/professional work	yes/no
Digital maturity index	Mean frequency of use of digital applications	0–4, ≥15 valid items, z-standardised
Digital competence	Self-rated digital competence score	continuous
Employer AI provision	Availability of AI applications provided by employer	yes/no/unclear
Prior AI training	Previous AI-specific education	yes/no

Supplementary Note 4

Free-text recoding of open-ended responses
Three multiple-response items offered an open-text option in addition to predefined categories: areas of AI use (Fig. 1b), formats of prior AI-specific training (Fig. 1c), and barriers to AI use (Fig. 1d). Predefined options were retained verbatim; free-text entries were mapped to an existing category where semantically equivalent, or marked uncodable (–). The authoritative mappings are implemented in results_R2.py.

Areas of AI use – recoding of free-text entries
Item: „Für welche der folgenden Bereiche nutzen Sie KI-Anwendungen? (Mehrfachnennungen möglich)“ Predefined categories (verbatim source in brackets): Administration [Administrative Aufgaben], Diagnostics [Diagnostik und Befund], Treatment planning [Behandlungsplanung und Therapieentscheidungen], Research [Forschung], Patient communication [Patientenkommunikation und Service].

Free-text entry (German)	Assigned category
Recherche	Research
Recherche, Patienteninformation	Research + Patient communication
Übersetzungen für fremdsprachliche Patienten	Patient communication
Spracherkennung und Literaturrecherche	Research + Administration
Behandlungsdokumentation	Administration
OpenEvidence	Research
Diktiergerät	Administration
Marketing	Administration
Dokumentation / Spracherkennung	Administration
Unterstützung bei seltenen Diagnosen …	Diagnostics
Bei der Strahlentherapie selbst	Diagnostics
zusätzliche Auswertung der Lunge	Diagnostics
Erstellen von Prüfungsfragen	– (uncodable)
Ich benutze meinen privaten Account	– (uncodable)
Nur wenig Nützliches … für Dienstplan etc. …	– (uncodable)

Nine of the 192 users contributed no codable area (seven left the field blank; two gave uncodable free text). They remain in the denominator (n = 192).
Table S4-2. Formats of prior AI-specific training – recoding
Item: „Welche Art von KI-Schulung war das? (Mehrfachnennung möglich)“ (trained physicians, n = 62).

Response (German)	Category
Kurzvortrag/Informationsveranstaltung (<2 Stunden)	Short lecture (<2 h)
Online-Modul/E-learning	E-learning module

Response (German)	Category
Workshop (wenige Stunden bis 1 Tag)	Workshop
Integrierte in anderen Fortbildungen	Integrated in other CME
Mehrtägiger Kurs/Zertifikatslehrgang	Certificate / multi-day
„Seminar des Hartmannbund … nach 15 min abgebrochen“	Short lecture (<2 h)
Keine Antwort	– (uncodable)

Barriers to AI use – recoding
Item: „Was sind die Hauptgründe, warum Sie derzeit keine KI-basierten Anwendungen nutzen?“ (definite non-users, n = 149).

Response / free-text entry (German)	Category
Nicht verfügbar	Not available
Fehlendes Wissen/Schulungen	Lack of knowledge / training
Sicherheits-/Datenschutzbedenken	Data protection / security concerns
Mangelnde Integrationsmöglichkeiten in bestehenden Systemen	Insufficient system integration
Keine wahrgenommene Nützlichkeit/Zeitgewinn	No perceived benefit
fehlende Kapazitäten / Zeitmangel für Implementierung	Lack of time / capacity
kein bedarf	No perceived benefit
„Als sporadisch in der Notfallpraxis tätiger Arzt … wird auch nicht bereitgestellt“	Not available
„Die Firmen … haben KI noch nicht … implementiert“	Not available
Kosten / „Kh hat kein Geld“	Costs
Testphase	Other

Costs and Other were each named by fewer than five respondents and are therefore not shown in Fig. 1d (see its caption).

Variables and coding for the regression, indices, and latent class analysis
Variables entered into the multivariable logistic regression (Table 2), the digital-maturity indices (Table 2; Supplementary Note 3), and the latent class analysis (Fig. 4; Supplementary Note 8) were derived from the questionnaire as follows.
Table S4-4. Variable definitions and coding

Variable	Questionnaire item (German)	Coding
AI use (outcome)	Nutzen Sie bereits KI-basierte Anwendungen in Ihrem beruflichen Alltag?	1 = any affirmative (regularly / occasionally / rarely); 0 = „Nein …“; „weiß nicht“ / no answer = missing (excluded from defined-use analyses, n = 341)

Variable	Questionnaire item (German)	Coding
Adoption segment	(same item)	user · interested non-user („Nein, aber geplant/Interesse vorhanden“) · no current intention („Nein, und aktuell nicht geplant“)
Age (age4)	Wie alt sind Sie?	<40 [ref] (20–29, 30–39) · 40–49 · 50–59 · 60+ (60–69, ≥70)
Sex (male)	Welchem Geschlecht ordnen Sie sich zu?	1 = männlich; 0 = weiblich [ref]
Digital competence	Wie schätzen Sie Ihre digitale Kompetenz im beruflichen Kontext ein?	continuous, 1–5
Prior AI training (trained)	Hatten Sie im beruflichen Kontext bereits Schulungen spezifisch zum Thema KI erhalten?	1 = Ja; 0 = Nein [ref]
Sector (ambulatory)	In welchem Umfeld arbeiten Sie hauptsächlich?	1 = ambulatory (Einzelpraxis, Gemeinschaftspraxis/BAG, MVZ); 0 = hospital [ref] (stationär, ambulant); public-health service, emergency medical service, no answer = missing
Employer provision	Stellt Ihnen Ihr Arbeitgeber KI-gestützte Anwendungen zur Verfügung?	yes · no [ref] · don’t know
Digital-maturity index	19 application items (Table S4-5)	mean of item frequencies (0–4); ≥15 valid items required; remaining missing set to 0; z-standardised within the analysis sample
Sector-neutral index (sensitivity)	8 items marked * in Table S4-5	as digital-maturity index; ≥6 valid items
LCA indicators	19 application items (Table S4-5)	each collapsed to 3 levels: 0 = not available / never; 1 = rarely–monthly; 2 = weekly–daily; ≥15 valid items (n = 356)

Frequency coding of the 19 application items (6-point scale). täglich = 4; wöchentlich = 3; monatlich = 2; seltener = 1; „möchte ich, aber nicht verfügbar“ = 0; „unklar, was das ist“ = 0.

The 19 digital-application items (* = sector-neutral subset)

No.	Item (German)	Short label (Fig. 4a)	Neutral
1	Praxisverwaltungssysteme (PVS) / Krankenhausinformationssystem (KIS)	Practice/hospital IS	*
2	Elektronische Patientenakte (ePA)	Electronic patient record (ePA)	
3	Elektronische Medikationsliste (eML/eMP)	e-medication list	*
4	Elektronische Arbeitsunfähigkeit (eAU)	e-sick note (eAU)	
5	Bildbetrachtungssysteme (z.B. PACS)	PACS / imaging	
6	Laborinformationssysteme (LIS)	Laboratory IS	*
7	Differentialdiagnostik bzw. Clinical-Decision-Support (CDSS)	Clinical decision support	*
8	Digitale Gesundheitsanwendungen (DiGA)	DiGA	
9	Gesundheits-Apps für die Patientenversorgung	Health apps	
10	Telefon-Assistenzsysteme (z.B. Terminvergabe)	Phone assistance	
11	Videosprechstunde	Video consultation	
12	Tele-Visite (z.B. Intensivstation / Stroke-Unit / Pflegeheim)	Tele-rounds	*
13	Tele-/Home-Monitoring	Tele-/home monitoring	
14	Spracherkennung zur automatisierten Behandlungs-Dokumentation	Speech recognition	
15	Digitale Patientenvoranmeldung (NIDA, IVENA, Pulsation)	Digital pre-registration	
16	Ersteinschätzungs-Tools (docyet, ada, SMeD)	Triage tools	*
17	Arzneimittel-Therapiesicherheitssystem (AMTS)	Medication safety (AMTS)	*
18	Automatisierte Dienstplanung	Automated duty scheduling	*
19	Wissensmanagement (z.B. Amboss, Elsevier, Thieme)	Knowledge management	

Correction rules for logically inconsistent records

During data cleaning (Supplementary Note 1), a logical-consistency check was applied to the employer-provision items. Records in which a respondent indicated that their employer provides AI applications (or named a form of provision) while

simultaneously reporting that no tools were available, or that they used only freely available („shadow“) AI, were treated as inconsistent.
Predefined rule: employer provision was recoded to „no“ and the associated „form of provision“ field was cleared. The rule was applied to three records (#89, #345, #316); all three remain in the analytic sample (n = 362).
This correction is distinct from the exclusion of records (Supplementary Note 1). One further record (#384) was excluded as a retired respondent and is not part of this correction.

Supplementary Note 5

Additional Information for RQ3, Age Group x Format

Item	<40	≥50
Self-paced online course	102/131 = 77.9%	112/171 = 65.5%
Information nuggets (15-30 min)	59/131 = 45.0%	50/171 = 29.2%

Supplementary Note 6

Segment comparison of all preference options: %, χ^2, Cramér's V, adjusted p-value + pairwise post-hoc tests (Fisher, BH) with effect sizes

block	dimension	option	pct_user	pct_interested	pct_no_intention	cramers_v	chi2	p	min_expected	p_adj	significant
W	Format	Self-paced online courses	78.125	73.2558	63.4921	0.1253	5.3576	0.0686	16.2581	0.1716	False
W	Format	Live webinars	56.25	54.6512	36.5079	0.1501	7.6865	0.0214	30.1144	0.0803	False
W	Format	Nuggets (15-30 min)	35.9375	44.186	36.5079	0.0726	1.7986	0.4069	24.0176	0.4988	False
W	Format	Blended learning	34.375	43.0233	31.746	0.0865	2.5531	279	22.7243	0.3986	False
W	Format	On-site sessions	33.3333	40.6977	31.746	0.0718	1.7556	0.4157	21.9853	0.4988	False
W	Scope	Short formats (1-2 h)	83.3333	79.0698	73.0159	0.0986	3.3181	0.1903	12.3783	0.3359	False
W	Scope	Half-day	23.4375	27.907	22.2222	0.0494	0.8325	0.6595	15.3343	0.7267	False
W	Scope	Longitudinal	18.75	25.5814	11.1111	0.1206	4.9627	0.0836	12.0088	193	False
W	Scope	Full-day	11.9792	11.6279	7.9365	0.0487	0.8102	0.6669	7.0205	0.7267	False
W	Scope	Multi-day	4.6875	6.9767	1.5873	0.0832	2.3618	307	2.956		False
W	Orientation	Application-oriented	91.1458	93.0233	88.8889	0.0477	0.7764	0.6783	5.5425	0.7267	False
C	Depth	Introductory	53.6458	51.1628	50.7937	0.0263	0.2363	0.8886	29.9296	0.8886	False
C	Depth	Specialised	50.0	23.2558	17.4603	0.3021	31.1161	0.0	23.4633	0.0	True
C	Depth	Strategic	38.5417	26.7442	17.4603	0.1799	11.0318	4	19.9531	0.0256	True
C	Topic	Diagnostic systems	47.9167	52.3256	28.5714	0.1656	9.3499	0.0093	28.6364	0.04	True
C	Topic	Treatment planning	44.2708	50.0	28.5714	0.1453	7.1999	0.0273	26.9736	0.0911	False
C	Topic	Practical implementation	41.1458	51.1628	31.746	0.1298	5.7426	0.0566	26.4194	0.1699	False
C	Topic	Large language models	41.6667	36.0465	17.4603	0.1884	12.1003	0.0024	22.5396	0.0256	True
C	Topic	Data protection	28.6458	23.2558	34.9206	0.0846	2.4395	0.2953	17.9208	0.4027	False

C	Topic	Diagnostic imaging	26.5625	22.093	15.873	0.0959	3.1383	0.2082	14.7801	347	False
C	Topic	Critical appraisal	19.2708	10.4651	31.746	0.1759	10.5523	0.0051	12.1935	0.0256	True
C	Topic	Ethics and law	14.0625	10.4651	30.1587	0.1861	11.8124	0.0027	10.1613	0.0256	True
C	Topic	Patient communication	11.9792	20.9302	11.1111	0.1148	4.4964	0.1056	8.868	0.2112	False
C	Topic	AI in medical research	4.1667	1.1628	1.5873	0.0834	2.3741	0.3051	1.8475		False
P	Provider	Chambers of physicians	61.9792	72.093	65.0794	0.0886	2.6742	0.2626	21.9853	0.3939	False
P	Provider	Professional associations	58.3333	60.4651	44.4444	0.1152	4.5283	0.1039	27.5279	0.2112	False
P	Provider	Employers	52.0833	47.6744	52.381	39	0.5188	0.7715	30.8534	0.7981	False
P	Provider	Specialty societies	53.125	46.5116	36.5079	0.1259	5.4067	67	30.4839	0.1716	False
P	Provider	Health insurance assoc.	30.2083	41.8605	38.0952	108	3.9798	0.1367	21.8006	0.2563	False
P	Provider	Universities	32.8125	26.7442	23.8095	0.0821	2.3002	0.3166	18.6598	413	False
P	Provider	Commercial AI vendors	23.4375	22.093	4.7619	0.1789	10.9148	0.0043	12.3783	0.0256	True
P	Provider	Private providers	16.1458	12.7907	7.9365	0.0904	2.7855	0.2484	8.6833	0.3922	False

Plus the extend Version mit Post-Tests

block	dimension	option	pct_user	pct_interested	pct_no_intention	n_user_named	n_interested_named	n_nointention_named
C	Depth	Specialised	50.0	23.2558	17.4603	96	20	11
C	Depth	Strategic	38.5417	26.7442	17.4603	74	23	11
C	Topic	Diagnostic systems	47.9167	52.3256	28.5714	92	45	18
C	Topic	Large language models	41.6667	36.0465	17.4603	80	31	11
C	Topic	Critical appraisal	19.2708	10.4651	31.746	37	9	20
C	Topic	Ethics and law	14.0625	10.4651	30.1587	27	9	19
P	Provider	Commercial AI vendors	23.4375	22.093	4.7619	45	19	3

chi2	p_omnibus	p_omnibus_adj	cramers_v	rd_pp_UvsI	ci95_low_UvsI	ci95_high_UvsI	p_pairwise_UvsI	p_pairwise_UvsI_adj	rd_pp_UvsN
31.1161	0.0	0.0	0.3021	26.7	14.6	37.1	0.0000	0.0003	32.5
11.0318	4	0.0256	0.1799	11.8	-0.3	22.6	0.0584	0.0876	21.1
9.3499	0.0093	0.04	0.1656	-4.4	-16.8	8.2	0.5186	0.5445	19.3
12.1003	0.0024	0.0256	0.1884	5.6	-6.9	17.3	0.4274	0.4943	24.2
10.5523	0.0051	0.0256	0.1759	8.8	-0.8	16.7	0.0810	0.1133	-12.5
11.8124	0.0027	0.0256	0.1861	3.6	-5.7	11.0	0.4472	0.4943	-16.1
10.9148	0.0043	0.0256	0.1789	1.3	-9.9	11.2	0.8781	0.8781	18.7

ci95_low_UvsN	ci95_high_UvsN	p_pairwise_UvsN	p_pairwise_UvsN_adj	rd_pp_IvsN	ci95_low_IvsN	ci95_high_IvsN	p_pairwise_IvsN	p_pairwise_IvsN_adj
19.4	42.7	0.0000	0.0001	5.8	-7.7	18.2	0.4212	0.4943
8.1	31.3	0.0020	0.0069	9.3	-4.6	21.9	0.2363	0.3102
5.4	31.3	0.0082	0.0157	23.8	7.8	37.8	0.0044	0.0103
11.2	34.5	0.0004	0.0031	18.6	4.0	31.5	0.0163	0.0285
-25.7	-0.6	0.0542	0.0876	-21.3	-34.5	-8.2	0.0016	0.0066
-29.0	-4.7	0.0074	0.0155	-19.7	-32.8	-6.8	0.0030	0.0089
8.7	25.9	0.0006	0.0034	17.3	6.1	27.7	0.0042	0.0103

pairwise_summary
users > interested (p_adj=0.000); users > no-intention (p_adj=0.000); interested = no-intention (ns)
users = interested (ns); users > no-intention (p_adj=0.007); interested = no-intention (ns)
users = interested (ns); users > no-intention (p_adj=0.016); interested > no-intention (p_adj=0.010)
users = interested (ns); users > no-intention (p_adj=0.003); interested > no-intention (p_adj=0.029)
users = interested (ns); users = no-intention (ns); interested < no-intention (p_adj=0.007)
users = interested (ns); users < no-intention (p_adj=0.016); interested < no-intention (p_adj=0.009)
users = interested (ns); users > no-intention (p_adj=0.003); interested > no-intention (p_adj=0.010)

Supplementary Note 7

Questionnaire first in German (original) then in English (translated).

Fragebogen (Original, Deutsch)

Studie zu Schulungsangeboten zu KI im Gesundheitswesen Vielen Dank, dass Sie an der Studie teilnehmen. (Microsoft Forms; * = Pflichtfeld. Originalschreibweise beibehalten, inkl. Tippfehler.)

Teil 1: Zustimmung zum Datenschutz und Einwilligung zur Studie

Vielen Dank, dass Sie an der Studie „Sensibilisierungs- und Schulungsangebote bezüglich Digitalisierung und Künstlicher Intelligenz (KI) innerhalb der deutschen Gesundheitsversorgung" teilnehmen.

Die Studie besteht aus drei Abschnitten: 1. Zustimmung zum Datenschutz und Einwilligung zur Studie; 2. Sozio-demographische Angaben; 3. Angaben zu bisherigen Digitalisierungs- und KI-Erfahrungen sowie Schulungen. Insgesamt werden für die Beantwortung der Fragen ca. 8–10 min benötigt. Bei weiteren Fragen oder Unklarheiten können Sie sich jederzeit an den Studienleiter Prof. Dr. Markus Schinle (markus.schinle@hs-offenburg.de) wenden.

1. Datenschutzerklärung zur Studie * (Datenschutzerklärung verlinkt.) Einwilligungstext: „Ich habe die Datenschutzerklärung … gelesen und willige die damit verbundene Datenverarbeitung ein. Mir ist bewusst, dass die Einwilligung freiwillig ist und ohne Nachteile verweigert oder jederzeit auch ohne Angaben von Gründen widerrufen werden kann. …"

- ☐ Ich habe die Datenschutzerklärung gelesen und stimme der Datenschutzerklärung zu.

2. Teilnehmendeninformationen zur Studie * (Teilnehmendeninformation verlinkt.) Einwilligungstext wie oben (Teilnahme + Datenverarbeitung).

- ☐ Ich habe die Teilnehmendeninformation gelesen und willige ein an der Studie teilzunehmen.

Teil 2: Sozio-demographische Fragen

3. Wie alt sind Sie? * 20–29 Jahre · 30–39 Jahre · 40–49 Jahre · 50–59 Jahre · 60–69 Jahre · 70 Jahre oder älter · keine Angabe

4. Welches Bundesland sind Sie tätig? * Baden-Württemberg · Bayern · Brandenburg · Berlin · Bremen · Hamburg · Hessen · Mecklenburg-Vorpommern · Niedersachsen · Nordrhein-Westfalen · Rheinland-Pfalz · Saarland · Sachsen · Sachen-Anhalt · Schleswig-Holstein · Thüringen · keine Angabe · Sonstiges

5. Welchem Geschlecht ordnen Sie sich zu? * männlich · weiblich · divers · keine Angabe

6. In welchem Umfeld arbeiten Sie hauptsächlich? * Krankenhaus (stationär) · Krankenhaus (ambulant) · Praxis (Einzelpraxis) · Praxis (Gemeinschaftspraxis/Berufsausübungsgemeinschaft) · MVZ (Medizinisches Versorgungszentrum) · Pflegeheim/Ambulante Pflege · Rettungsdienst · Öffentlicher Gesundheitsdienst · Verwaltung/Management · Selbstverwaltung/Politik/Berufsverband · keine Angabe

7. Welcher Berufsgruppe ordnen sie sich zu? * Medizinische Fachangestellte (MFA) · Professionelle Pflege · Ärzt:in · andere Healthcare Professionals · Sonstiges

8. In welcher Fachrichtung arbeiten Sie? * Allgemeinmedizin · Allgemeinchirurgie · Anästhestiologie · Arbeitsmedizin · Augenheilkunde · Dermatologie · Gastrologie · Gynäkologie · Geriatrie · Hals-Nasen-Ohrenheilkunde · Hämatologie und Onkologie · Humangenetik · Hygiene- und Umweltmedizin · Innere Medizin · Intensivmedizin · Kardiologie · Kinder- und Jugendmedizin · Kinder- und Jugendpsychiatrie und -psychotherapie · Labormedizin · Mund- und Kieferchirugie · Neurochirugie · Neurologie · Nuklearmedizin · Öffentliches Gesundheitswesen · Orthopädie und Unfallchirurgie (inkl. plastische und rekonstruktive Chirurgie) · Pathologie · Pharamakologie · Psychiatrie · Psychosomatik · Radiologie · Rechtsmedizin · Strahlenmedizin · Thorax-Herz-Gefäßchirurgie · Transfusionsmedizin · Urologie · keine Angaben · Sonstiges

9. Wie viel Berufserfahrung haben Sie? (Gerechnet ab dem dritten Staatsexamen) * *(Ärzt:innen)* 0–4 Jahre · 5–9 Jahre · 10–14 Jahre · 15–19 Jahre · 20–25 Jahre · >25 Jahre · kein Angabe

10. Was ist ihre Berufsbezeichnung/Position? (Pflege und MFA) * *(nur nicht-ärztliche Berufsgruppen)* Altenpfleger:in · Auszubildende:r zur/zum Medizinischen Fachangestellten (MFA) · Auszubildende:r in der Pflege · Medizinische/r Fachangestellte:r (MFA)/Arzthelfer:in · Leitende/r MFA · MFA mit Zusatzqualifikation (z.B. VERAH, NäPa, Praxismanager:in, digi Manager:in) · Notfallsanitäter:in bzw. Mitarbeiter:in im Rettungsdienst · Pflegeassistent:in bzw. Pflegehelfer:in · Gesundheits- und Krankenpfleger:in, Pflegefachfrau/mann · Praxisanleiter:in in der Pflege · Pflegefachkraft mit Weiterbildung (z.B. Intensivpflege, Anästhesie, OP, Kinderintensiv) · Physician Assistant · Stationsleitung bzw. Teamleitung · Pflegedienstleitung · Sonstiges

11. Wie viel Berufserfahrung haben Sie? (Gerechnet ab dem Abschluss der Ausbildung/Studium) * *(nicht-ärztliche Berufsgruppen)* 0–4 Jahre · 5–9 Jahre · 10–14 Jahre · 15–19 Jahre · 20–25 Jahre · >25 Jahre · kein Angabe

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Teil 3: Fragen zu Digitalisierungs- und KI-Erfahrungen sowie diesbezüglicher Schulungen

KI-Definition (Vorbemerkung): „Eine einheitliche Definition für Künstliche Intelligenz gibt es nicht. Daher möchten wir uns für diesen Fragebogen auf die Definition des Deutschen Ärzteblatts stützen: ‚dass es sich um ein Teilgebiet der Informatik sowie die Fähigkeit von Systemen handelt, selbstständig kognitive Funktionen wie Lernen, Denken, Planen und Problemlösung zu erfüllen, für die normalerweise menschliche Intelligenz nötig ist. Dabei unterscheidet sie sich (noch) in zentralen Eigenschaften wie dem Bewusstsein von menschlicher Intelligenz.'" (Deutsches Ärzteblatt, Sonderausgabe KI 2025)

12. Wie schätzen Sie Ihre digitale Kompetenz im Alltag (nicht beruflicher Kontext) ein? (1 = sehr niedrig … 5 = sehr hoch) 1 · 2 · 3 · 4 · 5

13. Wie häufig nutzen Sie folgende digitale Werkzeuge in Ihrem Berufsalltag? *(Matrix)* Skala: täglich · wöchentlich · monatlich · seltener · möchte ich, aber nicht verfügbar · unklar, was das ist Items (19): Praxisverwaltungssysteme (PVS)/Krankenhausinformationssystem (KIS) · Elektronische Patientenakte (ePA) · Elektronische Medikationsliste (eML/eMP) · Elektronische Arbeitsunfähigkeit (eAU) · Bildbetrachtungssysteme (z.B. PACS) · Laborinformationssysteme (LIS) · Differentialdiagnostik bzw. Clinical-Decision-Support (CDSS) · Digitale Gesundheitsanwendungen (DiGA) · Gesundheits-Apps für die Patientenversorgung · Telefon-Assistenzsysteme (z.B. für Terminvergabe) · Videosprechstunde · Tele-Visite (z.B. Intensivstation/Stroke-Unit/Pflegeheim) · Tele-/Home-Monitoring · Spracherkennung zur automatisierten Behandlungs-Dokumentation · Digitale Patientenvoranmeldung (NIDA, IVENA, Pulsation) in Notaufnahme/im Rettungsdienst · Ersteinschätzungs-Tools (wie z.B. von docyet, ada, SMeD) · Arzneimittel Therapiesicherheitssystem (AMTS) – Hinweis bei Medikamenten Anordnung · Automatisierte Dienstplanung · Wissensmanagement (z.B. Amboss, Elsevier, Thieme)

14. Stellt Ihnen Ihr Arbeitgeber KI-gestütze Anwendungen zur Verfügung? * Ja · Nein · Weiß ich nicht / Kann ich nicht einschätzen

15. In welcher Form werden diese KI-Tools bereitgestellt? * Als eigenständige Tools oder Programme (z.B. KI-Modelle oder Chatbots, Spracherkennung, usw.) · In bestehender medizinischer Software integriert (z.B. Diagnosesysteme, Entscheidungsunterstützung) · In Erprobung/Pilotphase (von externer Software) · Inhouse entwickelte KI-Modelle · Keine KI-Tools verfügbar · Weiß nicht · Sonstiges

16. Was sind die Hauptgründe, warum Sie derzeit keine KI-basierten Anwendungen nutzen? * — *Wählen Sie höchstens 3 Optionen aus.* Nicht verfügbar

· Sicherheits-/Datenschutzbedenken · Fehlendes Wissen/Schulungen · Keine wahrgenommene Nützlichkeit/Zeitgewinn · Mangelende Integrationsmöglichkeiten in bestehenden Systemen · fehlende Kapazitäten/Zeitmangel für Implementierung · Sonstiges

17. Wie schätzen Sie Ihre digitale Kompetenz im beruflichen Kontext ein? (1 = sehr niedrig … 5 = sehr hoch) * 1 · 2 · 3 · 4 · 5

18. Wie bewerten Sie die Unterstützung (z.B. Schulungen, Anleitungen) bei der Einführung neuer digitaler Werkzeuge an Ihrem Arbeitsplatz? * — *Freitext*

19. Nutzen Sie bereits KI-basierte Anwendungen in Ihrem beruflichen Alltag * Ja, regelmäßig (mehrfach pro Woche) · Ja, gelegentlich (einmal pro Woche) · Ja, selten/experimentell (einmal pro Monat) · Nein, aber geplant/Interesse vorhanden · Nein, und aktuell nicht geplant · Ich weiß es nicht/Bin mir unsicher · Keine Antwort

20. Für welche der folgenden Bereiche nutzen Sie KI-Anwendungen? (Mehrfachnennungen möglich) * Diagnostik und Befund (z.B. Bildanalyse, Mustererkennung, EKG-Auswertung etc.) · Behandlungsplanung und Therapieentscheidungen (z.B. Mediaktionsmanagement, personalisierte Therapieempfehlungen) · Administrative Aufgaben (z.B. Dokumentation wie Arztbrief schreiben, Abrechnung, etc.) · Forschung (z.B. Datenanalyse, Literaturrecherche, Simulationen) · Patientenkommunikation und Service (z.B. Chatbots, Terminvereinbarung, automatisierte Anamnese-Tools) · keine Antwort · Sonstiges

21. Wie setzen Sie sich im privaten Umfeld mit dem Thema KI auseinander? (Mehrfachnennung möglich) * Ich informiere mich aktiv (z.B. Fachartikel, Kurse, Podcasts) · Ich lese gelegentlich Nachrichten/Artikel dazu · Ich nutze KI-Anwendungen privat (z.B. ChatGPT, KI-Bildgeneratoren) · Über soziale Medien oder Netzwerke (z.B. LinkedIn) · Eher gar nicht/nur passiv · Keine Antwort · Sonstiges

22. Hatten Sie im beruflichen Kontext bereits Schulungen spezifisch zum Thema KI erhalten? * Ja · Nein · Keine Antwort

23. Welche Art von KI-Schulung war das? (Mehrfachnennung möglich) * Kurzvortrag/Informationsveranstaltung (<2 Stunden) · Workshop (wenige Stunden bis 1 Tag) · Mehrtägiger Kurs/Zertifikatslehrgang · Online-Modul/E-learning · Integrierte in anderen Fortbildungen · Keine Antwort · Sonstiges

24. Wie bewerten Sie die Nützlichkeit dieser Schulung(en) für Ihre Arbeit? (1 = gar nicht brauchbar … 5 = sehr nützlich) * 1 · 2 · 3 · 4 · 5

25. Welche Formate für KI-Schulungen würden Sie bevorzugen? (Mehrfachnennungen möglich) * Online-Kurse (flexibel, Selbstlernformat) · Live Online-Webinare (interaktiv) · Präsenzveranstaltungen/Workshop vor Ort · Blended Learning (Kombination aus Online und Präsenz) · Kurze Informations-Nuggets (z.B. 15–30 min Videos) · Kein Präferenz · Sonstiges

26. Welchen Umfang/Dauer für KI-Schulungen fänden Sie ideal (Mehrfachnennung möglich)? * Kurzformate (z.B. 1–2 Stunden) · Halbtägige

Veranstaltungen · Ganztägige Veranstaltungen · Mehrtägige Kurse · Längerfristige Weiterbildungen (z.B. über mehrere Wochen/Monate verteilt) · Keine Präferenz · Sonstiges

27. Auf welchem Niveau sollten die KI-Schulungen für Sie idealerweise ansetzen? (Mehrfachnennung möglich) * Einführend/Grundlagen vermittelnd (Fokus auf Verständnis der Basiskonzepte) · Anwendungsorientiert/Praxisbezogen (Fokus auf Nutzung spezifischer Tools und Methoden im Alltag) · Vertiefend/Spezialisiert (Fokus auf detaillierte Kenntnisse spezifischer KI-Methoden oder komplexer Anwendungsfälle) · Strategisch/Managementbezogen (Fokus auf Implementierung) · Keine Antwort · Sonstiges

28. Welche der folgenden Themenbereiche für KI-Schulungen wären für Sie besonders relevant? * — *Wählen Sie höchstens 3 Optionen aus.* KI in der diagnostischen Bildgebung · KI-gestützte Diagnosesysteme · KI in der Therapieplanung und personalisierter Medizin · Umgang mit großen Sprachmodellen (z.B. ChatGPT) im medizinischen Kontext · Datenmanagement und Datenschutz bei KI-Anwendungen · Ethische und rechtliche Rahmenbedigungen für KI in der Medizin · Kritische Berwertung von KI-Anwendungen und Studien · Praktische Implementierung von KI-Tools im Arbeitsalltag · KI in der medizinischen Forschung · KI und Patienteninteraktion und -kommunikation · Keine Antwort · Sonstiges

29. Welche Organisation oder Institution sollte dieses Angebot anbieten (Mehrfachnennung möglich)? * Ärztekammern · Arbeitgeber (Krankenhäuser/Arztpraxen) · Berufsverbände (z.B. Hartmannbund e.V., Marburger Bund) · Fachgesellschaften (z.B. DGIM, DEGAM, DGOU) · Industrieanbieter von KI-Software · Kassenärztliche Vereinigungen · Unabhängige private Weiterbildungsinstitute · Universitäten/Hochschulen · Keine Präferenz · Sonstiges

30. Welche Gründe halten Sie davon ab, sich zu KI in der Medizin weiterzubilden? * — *Freitext*

31. Haben Sie weitere Anmerkungen oder Wünsche bezüglich KI-Schulungen für Ihre Berufsgruppe? * — *Freitext*

32. Kennen Sie Fälle von Schatten-KI in ihrem beruflichen Umfeld? * *Definition:* „Unter Schatten-KI versteht man die Verwendung von KI-basierten Anwendungen oder Tools durch Ärzt:innen, die ohne offizielle Genehmigung oder ohne Aufsicht durch die zuständigen IT- oder Datenschutzverantwortlichen erfolgt. Beispiel: Die Nutzung von ChatGPT auf dem privaten Endgerät zur Diagnose-Stellung." Ja, habe selbst schon mal KI ohne formelle Genehmigung genutzt. · Ja, habe andere schon mal KI ohne formelle Genehmigung nutzen sehen. · Nein, mir ist kein Fall bekannt · keine Antwort

33. Haben Sie von ihrem Arbeitgeber eine Leitlinie oder ähnliches zum Einsatz und der Nutzung von Künstlicher Intelligenz, insbesondere Large Language Models (ChatGPT, Microsoft Co-Pilot, Claude, etc.) erhalten? Ja · Nein · Weiß ich nicht · Sonstiges

Questionnaire in English (translated)
This is an English translation of the original German questionnaire administered via Microsoft Forms. Item numbering follows the original. Question type and any selection limits are reproduced as displayed.

Conditional display logic (branching) is not represented in the Microsoft Forms export and is therefore not captured here; see the notes on items 15, 16, 23 and 24. Spelling errors present in the German original have been silently corrected in translation. The English wording of items reported in the manuscript follows the wording used there.

Part 1: Consent to data protection and participation in the study

Thank you for taking part in the study 'Awareness and training offerings regarding digitalisation and artificial intelligence (AI) within German healthcare provision'.

The study consists of three sections: (1) consent to data protection and participation in the study; (2) socio-demographic information; (3) information on previous digitalisation and AI experiences and related training.

Answering the questions takes approximately 8 to 10 minutes in total.

If you have further questions or uncertainties, you may contact the study leader [name and institutional email redacted] at any time.

Q1. I have read the data protection declaration and agree to it.

Single choice. Required. A link to the data protection declaration was provided in the original [link redacted].

Consent statement shown to respondents: I have read the data protection declaration for participation in the study and consent to the associated data processing. I am aware that consent is voluntary and may be refused without disadvantage or withdrawn at any time without giving reasons. I understand that, in the event of withdrawal, the lawfulness of the processing carried out on the basis of consent up to that point is not affected, and that to withdraw I can simply contact the person named in the participant information and the data protection declaration.

Q2. I have read the participant information and consent to take part in the study.

Single choice. Required. A link to the participant information was provided in the original [link redacted].

Consent statement shown to respondents: I have read the participant information and the data protection declaration for participation in the study and consent to participation and the associated data processing. The voluntariness and withdrawal conditions stated for item 1 apply equally.

Part 2: Socio-demographic questions

Q3. How old are you?

Single choice. Required.

- 20 to 29 years
- 30 to 39 years
- 40 to 49 years
- 50 to 59 years
- 60 to 69 years
- 70 years or older
- No answer

Q4. In which federal state do you work?

Single choice. Required.

- Baden-Wuerttemberg
- Bavaria
- Brandenburg
- Berlin
- Bremen
- Hamburg
- Hesse
- Mecklenburg-Western Pomerania
- Lower Saxony
- North Rhine-Westphalia
- Rhineland-Palatinate
- Saarland
- Saxony
- Saxony-Anhalt
- Schleswig-Holstein
- Thuringia
- No answer
- Other

Q5. Which gender do you identify with?

Single choice. Required.

- Male
- Female
- Diverse
- No answer

Q6. In which setting do you mainly work?

Single choice. Required.

- Hospital (inpatient)
- Hospital (outpatient)
- Practice (single practice)
- Practice (group practice / joint medical practice)
- Medical care centre (MVZ)
- Nursing home / outpatient care
- Emergency medical service
- Public health service
- Administration / management
- Self-governance / politics / professional association
- No answer

Q7. Which professional group do you belong to?

Single choice. Required.

- Medical assistant (MFA)
- Professional nursing
- Physician
- Other healthcare professionals
- Other

Q8. In which specialty do you work?

Single choice. Required.

- General medicine
- General surgery
- Anaesthesiology
- Occupational medicine
- Ophthalmology
- Dermatology
- Gastroenterology
- Gynaecology
- Geriatrics
- Otorhinolaryngology (ENT)
- Haematology and oncology
- Human genetics
- Hygiene and environmental medicine
- Internal medicine
- Intensive care medicine
- Cardiology
- Paediatric and adolescent medicine
- Child and adolescent psychiatry and psychotherapy
- Laboratory medicine
- Oral and maxillofacial surgery
- Neurosurgery
- Neurology
- Nuclear medicine
- Public health
- Orthopaedics and trauma surgery (incl. plastic and reconstructive surgery)
- Pathology
- Pharmacology
- Psychiatry
- Psychosomatic medicine
- Radiology
- Forensic medicine
- Radiation medicine
- Thoracic, cardiac and vascular surgery
- Transfusion medicine
- Urology
- No answer
- Other

Q9. How much professional experience do you have? (Counted from the third state examination)

Single choice. Required. The reference point indicates that this item is addressed to physicians.

- 0 to 4 years
- 5 to 9 years
- 10 to 14 years
- 15 to 19 years
- 20 to 25 years
- More than 25 years
- No answer

Q10. What is your job title / position? (Nursing and medical assistants)

Single choice. Required. Addressed to nursing staff and medical assistants.

- Geriatric nurse
- Trainee medical assistant (MFA)
- Nursing trainee
- Medical assistant (MFA) / doctor's assistant
- Lead medical assistant
- Medical assistant with additional qualification (e.g. VERAH, NaePa, practice manager, digital manager)
- Paramedic or emergency medical service staff
- Nursing assistant or nursing aide
- Registered nurse (health and nursing care professional)
- Practice instructor in nursing
- Specialist nurse with further training (e.g. intensive care, anaesthesia, operating theatre, paediatric intensive care)
- Physician assistant
- Ward manager or team leader
- Nursing service manager
- Other

Q11. How much professional experience do you have? (Counted from completion of training / studies)

Single choice. Required. Addressed to nursing staff and medical assistants.

- 0 to 4 years
- 5 to 9 years
- 10 to 14 years
- 15 to 19 years
- 20 to 25 years
- More than 25 years
- No answer

Part 3: Questions on digitalisation and AI experiences and related training

Definition shown to respondents: There is no uniform definition of artificial intelligence. For this questionnaire we therefore rely on the definition of the Deutsches Aerzteblatt, namely that AI is a subfield of computer science and the ability of systems to perform independently cognitive functions such as learning, thinking, planning and problem-solving that normally require human intelligence, while (still) differing from human intelligence in central characteristics such as consciousness. Source: Deutsches Aerzteblatt, AI special issue 2025.

Q12. How would you rate your digital competence in everyday life (non-professional context)?

Rating scale 1 to 5, where 1 = very low and 5 = very high.

Q13. How frequently do you use the following digital tools in your working day?

Matrix question. One frequency selected per row. Frequency categories: daily, weekly, monthly.

Tool	Daily	Weekly	Monthly
Practice management systems (PVS) / hospital information system (HIS)			
Electronic patient record (ePA)			
Electronic medication list (eML / eMP)			
Electronic certificate of incapacity for work (eAU)			
Image viewing systems (e.g. PACS)			
Laboratory information systems (LIS)			
Differential diagnostics / clinical decision support (CDSS)			
Digital health applications (DiGA)			
Health apps for patient care			
Telephone assistance systems (e.g. for appointment scheduling)			
Video consultation			
Tele-visit (e.g. intensive care unit / stroke unit / nursing home)			
Tele- / home monitoring			
Speech recognition for automated treatment documentation			
Digital patient pre-registration (NIDA, IVENA, Pulsation) in the emergency department / emergency medical service			
Initial assessment / triage tools (e.g. from docyet, ada, SMeD)			
Drug therapy safety system (AMTS): alerts when prescribing medication			
Automated duty / shift scheduling			
Knowledge management (e.g. Amboss, Elsevier, Thieme)			

Q14. Does your employer provide you with AI-based applications?

Single choice. Required. This item is the basis for the governance index component 'provision of AI tools' reported in the manuscript.

- Yes
- No
- I don't know / cannot say

Q15. In what form are these AI tools provided?

Multiple choice. Required. Conditional on item 14; displayed where AI tools are provided.

- As standalone tools or programs (e.g. AI models or chatbots, speech recognition, etc.)
- Integrated into existing medical software (e.g. diagnostic systems, decision support)
- In trial / pilot phase (external software)

- In-house developed AI models
- No AI tools available
- Don't know
- Other

Q16. What are the main reasons why you currently do not use AI-based applications?

Multiple choice, select at most 3. Required. Conditional; addressed to respondents who do not use AI applications.

- Not available
- Security / data protection concerns
- Lack of knowledge / training
- No perceived usefulness / time saving
- Lack of integration options in existing systems
- Lack of capacity / time for implementation
- Other

Q17. How would you rate your digital competence in a professional context?

Rating scale 1 to 5, where 1 = very low and 5 = very high. Required.

Q18. How do you rate the support (e.g. training, instructions) when introducing new digital tools at your workplace?

Free text. Required.

Q19. Do you already use AI-based applications in your professional daily routine?

Single choice. Required. Provides the AI-use frequency categories reported in the manuscript.

- Yes, regularly (several times per week)
- Yes, occasionally (once per week)
- Yes, rarely / experimentally (once per month)
- No, but planned / interested
- No, and not currently planned
- I don't know / am unsure
- No answer

Q20. For which of the following areas do you use AI applications? (Multiple answers possible)

Multiple choice. Required. Provides the application-area figures reported in the manuscript.

- Diagnostics and findings (e.g. image analysis, pattern recognition, ECG evaluation, etc.)
- Treatment planning and therapy decisions (e.g. medication management, personalised therapy recommendations)
- Administrative tasks (e.g. documentation such as writing physician letters, billing, etc.)
- Research (e.g. data analysis, literature search, simulations)
- Patient communication and service (e.g. chatbots, appointment scheduling, automated history-taking tools)
- No answer
- Other

Q21. How do you engage with the topic of AI in your private life? (Multiple answers possible)

Multiple choice. Required.

- I actively inform myself (e.g. specialist articles, courses, podcasts)
- I occasionally read news / articles about it
- I use AI applications privately (e.g. ChatGPT, AI image generators)
- Via social media or networks (e.g. LinkedIn)
- Hardly at all / only passively
- No answer
- Other

Q22. Have you already received training specifically on the topic of AI in a professional context?

Single choice. Required. This item is the basis for the governance index component 'AI-specific training'.

- Yes
- No
- No answer

Q23. What type of AI training was it? (Multiple answers possible)

Multiple choice. Required. Conditional on item 22; displayed where training was received.

- Short presentation / information session (under 2 hours)
- Workshop (a few hours up to 1 day)
- Multi-day course / certificate programme
- Online module / e-learning
- Integrated into other continuing education
- No answer
- Other

Q24. How do you rate the usefulness of this training for your work?

Rating scale 1 to 5, where 1 = not useful at all and 5 = very useful. Required. Conditional on item 22.

Q25. Which formats for AI training would you prefer? (Multiple answers possible)

Multiple choice. Required.

- Online courses (flexible, self-study format)
- Live online webinars (interactive)
- In-person events / on-site workshop
- Blended learning (combination of online and in-person)
- Short information nuggets (e.g. 15 to 30 minute videos)
- No preference
- Other

Q26. What scope / duration for AI training would you consider ideal? (Multiple answers possible)

Multiple choice. Required.

- Short formats (e.g. 1 to 2 hours)
- Half-day events
- Full-day events
- Multi-day courses
- Longer-term continuing education (e.g. spread over several weeks / months)

- No preference
- Other

Q27. At what level should the AI training ideally be pitched for you? (Multiple answers possible)

Multiple choice. Required.

- Introductory / foundational (focus on understanding basic concepts)
- Application-oriented / practical (focus on using specific tools and methods in daily work)
- Advanced / specialised (focus on detailed knowledge of specific AI methods or complex use cases)
- Strategic / management-oriented (focus on implementation)
- No answer
- Other

Q28. Which of the following topic areas for AI training would be particularly relevant for you?

Multiple choice, select at most 3. Required.

- AI in diagnostic imaging
- AI-supported diagnostic systems
- AI in therapy planning and personalised medicine
- Handling large language models (e.g. ChatGPT) in the medical context
- Data management and data protection in AI applications
- Ethical and legal frameworks for AI in medicine
- Critical appraisal of AI applications and studies
- Practical implementation of AI tools in daily work
- AI in medical research
- AI and patient interaction and communication
- No answer
- Other

Q29. Which organisation or institution should provide this offering? (Multiple answers possible)

Multiple choice. Required.

- Medical chambers (Aerztekammern)
- Employers (hospitals / medical practices)
- Professional associations (e.g. Hartmannbund e.V., Marburger Bund)
- Specialist societies (e.g. DGIM, DEGAM, DGOU)
- Industry providers of AI software
- Associations of Statutory Health Insurance Physicians (Kassenaerztliche Vereinigungen)
- Independent private continuing education institutes
- Universities / higher education institutions
- No preference
- Other

Q30. What reasons keep you from pursuing further training on AI in medicine?

Free text. Required.

Q31. Do you have any further comments or requests regarding AI training for your professional group?

Free text. Required.

Q32. Are you aware of cases of Shadow AI in your professional environment?

Single choice. Required. This is the core Shadow AI outcome item. The single-choice format does not allow respondents to report both own use and observation of others; the manuscript therefore treats the exposure rate as a lower bound.

Definition shown to respondents: Shadow AI refers to the use of AI-based applications or tools by physicians without official authorisation or without oversight by the responsible IT or data protection officers. Example: using ChatGPT on a private device to make a diagnosis.

- Yes, I have used AI myself without formal authorisation before.
- Yes, I have seen others use AI without formal authorisation before.
- No, I am not aware of any such case.
- No answer

Q33. Have you received a guideline or similar from your employer on the deployment and use of artificial intelligence, in particular large language models (ChatGPT, Microsoft Co-Pilot, Claude, etc.)?

Single choice. This item is the basis for the governance index component 'written guidelines'.

- Yes
- No
- I don't know
- Other

Supplementary Note 8

The primary analysis retained a three-class solution (Fig. 4). Although the Bayesian information criterion (BIC) reached its minimum at four classes, the additional class did not alter the substantive pattern: it split the digitally basic environment into two sector-specific application profiles. The four-class solution is therefore reported here as a sensitivity analysis. All models were estimated with StepMix (categorical LCA; 19 items on three levels; 20 random starts; random_state = 42; n = 356 with ≥15 valid items), identical to the primary analysis.

Model selection, k = 2–5

k	BIC	Entropy	Class sizes
2	11694	0.929	190, 166
3	11601	0.931	181, 113, 62
4	11550	0.915	111, 99, 85, 61
5	11588	0.921	92, 81, 80, 56, 47

BIC is minimal at k = 4; entropy is highest at k = 3. Lower BIC indicates better fit.

Four-class solution, class profiles

Class	n	Share %	Digital index (M)	AI use %	Provision %	Training %
1 Frontrunner	61	17.1	2.35	82.0	60.7	26.2
2 Intermediate	111	31.2	1.52	51.9	19.8	11.7
3 Basic (ambulatory)	85	23.9	1.43	42.9	18.8	21.2
4 Basic (hospital)	99	27.8	1.16	54.8	28.3	14.1

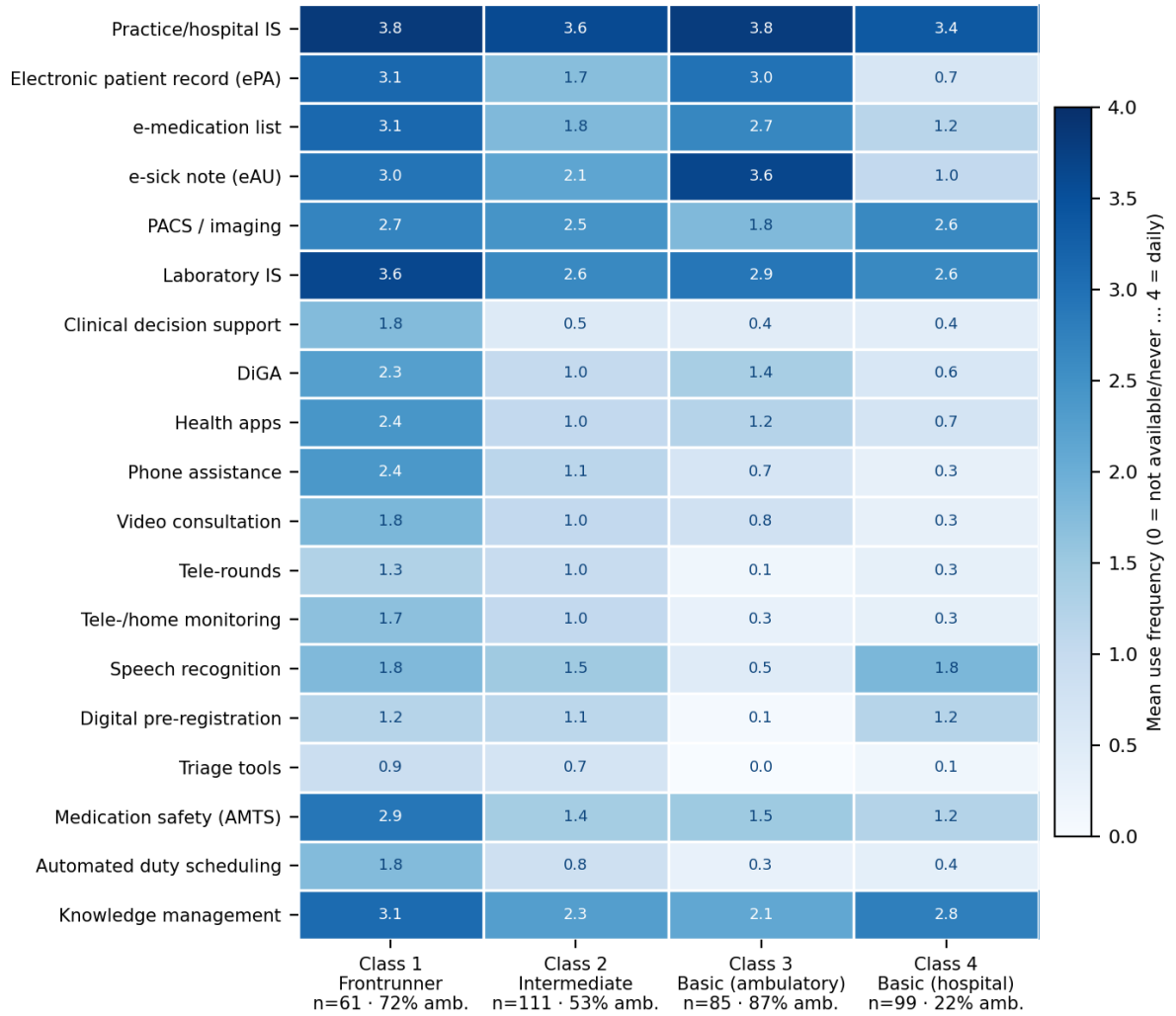
Classes ordered by digital-maturity index. χ^2 tests across the four classes: AI use $\chi^2(3) = 22.9$, $p < 0.001$ (defined-status respondents, n = 335); employer provision $\chi^2(3) = 38.6$, $p < 0.001$; prior AI training $\chi^2(3) = 7.5$, $p = 0.059$.

Correspondence with the three-class solution and sector composition

Four-class	Frontrunner	Intermediate	Basic	% ambulatory
1 Frontrunner	59	0	2	71.7
2 Intermediate	1	108	2	52.5
3 Basic (ambulatory)	2	2	81	87.1
4 Basic (hospital)	0	3	96	22.1

Cell entries are counts of respondents (four-class membership × three-class membership). Classes 1 and 2 reproduce the frontrunner and intermediate classes almost exactly; Classes 3 and 4 partition the basic class (n = 181) into an ambulatory-dominated (87% ambulatory) and a hospital-dominated (22% ambulatory) profile.

Mean use frequency of the 19 digital-application items across the four latent classes



Mean coded frequency (0 = not available/never … 4 = daily) per item and class. Class 3 (ambulatory-dominated) shows high use of ambulatory telematics applications (e-sick note 3.6, electronic patient record 3.0, e-medication list 2.7) but low imaging and hospital-specific tools; Class 4 (hospital-dominated) shows the reverse pattern (PACS 2.6, speech recognition 1.8, knowledge management 2.8). The two basic subclasses have comparable overall maturity (index 1.43 vs 1.16) but differ in application profile.

Conclusion

Across both solutions the substantive findings are unchanged: professional AI use remained close to one half in the basic subclasses (Class 3, 42.9%; Class 4, 54.8%) and prior AI-specific training was uniformly low across classes (11.7–26.2%; p = 0.059). Because the fourth class reproduces a sector distinction already captured by the sector variable and does not change the conclusions, the more parsimonious three-class solution (entropy 0.93) was retained as the primary model.

Supplementary Note 9

CHERRIES Checklist (Checklist for Reporting Results of Internet E-Surveys)

Reporting of this cross-sectional web-based survey follows the CHERRIES checklist. Items shaded in orange could not be reported, or are not applicable, given the anonymous convenience-sample design; these limitations are acknowledged in the manuscript.

Checklist item	CHERRIES description	Reported in this study (location)
1. Design		
Describe survey design	Describe target population, sample, whether the sample was self-selected, and the general design (e.g. cross-sectional).	Cross-sectional web-based survey; non-probability convenience sample of physicians practising in Germany (target population N ≈ 446,100). Self-selected respondents recruited via the Hartmannbund email distribution list. (Methods > Study design and participants; Recruitment.)
2. IRB (Institutional Review Board) approval and informed consent process		
IRB approval	Mention whether the study has been approved by an IRB/ethics committee.	Approved by University of Cambridge (Ref. 736, 15 May 2025) and Offenburg University of Applied Sciences (Ref. 04-25, 27 June 2025); conducted per the Declaration of Helsinki. (Methods > Ethics.)
Informed consent	Describe the informed consent process and where it was displayed.	A privacy notice and participant information sheet preceded the questionnaire; participation was conditional on informed consent from all participants. (Methods > Ethics.)
Data protection	State whether and how the survey conforms with data-protection regulations.	Data collected in compliance with the GDPR; participation was voluntary and anonymous, with no personally identifying data collected. (Methods > Ethics; Recruitment.)
3. Development and pre-testing		
Development and testing	State how the survey was developed, including pre-testing, usability and functionality.	Instrument newly developed (no validated instrument available); items drew on prior exploratory interviews and related projects. Piloted with 15 junior physicians (Arbeitskreis Junge Ärztinnen und Ärzte, Hartmannbund) who completed the full questionnaire and gave structured feedback on comprehensibility, terminology and response options; response categories were revised accordingly. Full instrument in Supplementary Note 7. (Methods > Instrument development.)
4. Recruitment process and description of the sample having access to the questionnaire		
Open survey vs closed survey	State whether the survey was open to any visitor, or a closed survey for a defined sample.	Closed survey: distributed to the defined membership of the Hartmannbund email distribution list; not publicly advertised or indexed. (Methods > Recruitment and participants.)
Contact mode	Indicate whether initial contact with potential participants was made on the Internet.	Initial contact made electronically by email via the central Hartmannbund distribution list. (Methods > Recruitment and participants.)
Advertising the survey	How/where was the survey announced or advertised?	Announced through a single email invitation (December 2025) with one email reminder (January 2026); no other advertising channels. (Methods > Recruitment and participants.)
5. Survey administration		

Web / E-mail	State whether it was a Web- or email-based survey.	Web-based survey administered via Microsoft Forms; the link was distributed by email. (Methods – Study design and participants.)
Context	Describe the Web site/page in which the survey was hosted.	Hosted on Microsoft Forms, optimised for desktop and mobile; the survey formed part of a larger project on digitalisation and AI adoption among physicians. (Methods > Study design and participants.)
Mandatory / voluntary	Was the survey mandatory or voluntary?	Voluntary and anonymous. (Methods > Recruitment and participants.)
Incentives	Were any incentives offered?	No incentives were offered. (Methods > Recruitment and participants.)
Time / Date	In which time frame was the survey conducted?	December 2025 to February 2026. (Methods > Study design and participants.)
Randomisation of items or questionnaires	To prevent bias, were items/pages randomised?	Not applicable. Items were presented in a fixed order; no randomisation was used.
Adaptive questioning	Were items dynamically displayed based on prior answers, to reduce complexity?	Adaptive/conditional branching was not used; all respondents saw the same items (the barriers item was answered by everyone but analysed only for definite non-users). (Methods > Measures.)
Number of items	What was the number of questionnaire items per page and in total?	Questionnaire comprised three thematic blocks (digitalisation/AI status quo; unsanctioned AI use, analysed elsewhere; training experience and preferences). Exact item count not stated in text; full questionnaire reproduced in Supplementary Note 7.
Number of screens (pages)	Over how many pages was the questionnaire distributed?	Number of screens/pages not reported. Administered via Microsoft Forms; median completion time ~8 to 10 minutes. (Methods > Instrument development.)
Completeness check	Was completeness checked before submission?	Not reported at the form level; missing values in key demographic/outcome variables were handled at the data-cleaning stage (Supplementary Note 1).
Review step	Could respondents review and change their answers (e.g. Back button)?	Not explicitly reported. Microsoft Forms permits back-navigation before final submission.
6. Response rates		
Unique site visitor	How were unique visitors determined (cookie, IP, log file)?	Not determinable. Anonymous email-list distribution with no site-visitor tracking; number of invitations delivered is not quantifiable. Acknowledged as a limitation. (Discussion > Limitations.)
View rate	Ratio of unique survey visitors / unique site visitors.	Not calculable (denominator unknown; self-selected sample without a quantifiable response rate). (Discussion > Limitations.)
Participation rate	Ratio of unique visitors who agreed to participate / unique first survey-page visitors.	Not calculable for the same reason.
Completion rate	Ratio of users who finished the survey / users who agreed to participate.	403 questionnaires were received; 41 excluded on predefined eligibility/data criteria, yielding 362 analysed. A completion rate relative to those who started cannot be computed. (Results > Participant characteristics; Supplementary Note 1.)

7. Preventing multiple entries from the same individual		
Cookies used	Were cookies used to assign a unique user identifier?	Not reported; the survey was anonymous.
IP check	Were IP addresses checked to prevent duplicates?	IP addresses were not collected (anonymous, GDPR-compliant); no IP-based duplicate check.
Log file analysis	Were other techniques used to analyse the log file for identical entries?	Raw responses were screened for duplicates during data cleaning; no duplicate entries were identified. (Supplementary Note 1.)
Registration	Was the survey open only to registered users identified by a code/password?	No registration or login; open to any recipient of the email invitation but confined to that closed distribution list.
8. Analysis		
Handling of incomplete questionnaires	Were incomplete questionnaires analysed?	Records missing key demographic/outcome variables were excluded (Supplementary Note 1); the regression used complete cases (n = 322). 21 participants with undefined AI-use status were excluded from segment analyses (n = 341). (Methods > Measures/Statistical analysis.)
Questionnaires submitted with an typical timestamp	Were surveys with atypical completion times identified/excluded?	No timestamp-based exclusion was applied or reported.
Statistical correction	Were weighting or propensity-score methods used to adjust for a non-representative sample?	No post-stratification weighting applied; representativeness was assessed by comparison with Bundesärztekammer statistics (Supplementary Note 6). Multiplicity controlled by Benjamini–Hochberg FDR correction; robustness checked in four sensitivity analyses (Supplementary Note 3). (Methods > Statistical analysis.)